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Quotient spaces

Example 1.0.1

Let $\mathbb{D}^n = \{x \in \mathbb{R}^n \mid \|x\| \leq 1\}$ be the n -disk and let $\mathbb{S}^{n-1} = \{x \in \mathbb{R}^n \mid \|x\| = 1\}$ be the $(n-1)$ -sphere. Then $\mathbb{D}^n/\mathbb{S}^{n-1}$ is homeomorphic to $\mathbb{S}^n$.

Proof. Given $x = (x_1, \dots, x_n) \in \mathbb{R}^n$, we let $i(x) = (x_1, \dots, x_n, 0) \in \mathbb{R}^{n+1}$. We let $u = (0, \dots, 0, 1) \in \mathbb{R}^{n+1}$. Then we define: $f : \mathbb{D}^n \rightarrow \mathbb{S}^n$, $f(x) := \cos(\pi\|x\|)u + \sin(\pi\|x\|)\frac{i(x)}{\|x\|}$. Then f is a continuous map which induces a continuous map $\bar{f} : \mathbb{D}^n/\mathbb{S}^{n-1} \rightarrow \mathbb{S}^n$. We can then check that $\bar{f}$ is a bijection with a compact source and Hausdorff target, thus it is a homeomorphism.

□

□

Fundamental Group

Covering Space

3.1 Basic Definitions

Definition 3.1.1 ► Covering Space

Let $p : \bar{X} \rightarrow X$ be a surjective map.

- We say an open subset V of X is **evenly covered** by p , if $p^{-1}(V)$ is a disjoint union of open subsets of $\bar{X}$:

$$p^{-1}(V) = \coprod_i U_i$$

where, each U_i is mapped homeomorphically onto V by p .

- In this case, we shall refer to each U_i as a **sheet** for p over V .
- If the space X can be covered by open subsets which are evenly covered by p , then we say that p is a **covering projection**; the space $\bar{X}$ is called a covering space of X .

Theorem 3.1.2

Let $p : \bar{X} \rightarrow X$ be a continuous map, where X is locally path connected.

1. The map p is a covering space iff for each component $\bar{C}$ of $\bar{X}$, the restriction map $p : p^{-1}(C) \rightarrow C$ is a covering projection.
2. If p is a covering projection then for each component $\bar{C}$ of $\bar{X}$, the map $p : \bar{C} \rightarrow p(\bar{C})$ is a covering projection and $p(\bar{C})$ is a component of X .

Remark.

Thus, we always assume that a covering space is locally path connected and connected. For the following content, unless otherwise specified, we will all adopt this assumption.

Proposition 3.1.3

Let $p : \bar{X} \rightarrow X$ be a covering projection. If X is Hausdorff then $\bar{X}$ is Hausdorff. Further, if p is finite-to-one, then show that if $\bar{X}$ is Hausdorff then X is Hausdorff.

Note.

I haven't finish the proof of the second part yet. Actually, if a covering projection is proper it is also closed. Once we have this conclusion we can then finish the proof easily.

Proof. 1. Take two different points $\bar{x}, \bar{y} \in \bar{X}$. Denote $x := p(\bar{x}), y := p(\bar{y})$.

- (a) If $x \neq y$: Since X is Hausdorff, there exists two open sets U, V such that $x \in U, y \in V, U \cap V = \emptyset$. We have $p^{-1}(U), p^{-1}(V)$ are two open subsets of $\bar{X}$ with $p^{-1}(U) \cap p^{-1}(V) = \emptyset$. Since $\bar{x} \in p^{-1}(U), \bar{y} \in p^{-1}(V)$, it shows that $\bar{X}$ is Hausdorff.
- (b) If $x = y$: Let W be a open neighbourhood of $x = y$. We write $p^{-1}(W) = \coprod_i W_i$. Since $\bar{x}, \bar{y} \in \coprod_i W_i$ and any two W_i are disjoint, there are two different W_{i_1}, W_{i_2} such that $\bar{x} \in W_{i_1}, \bar{y} \in W_{i_2}$.

The above shows that $\bar{X}$ is Hausdorff.

- 2. Take two different points $x, y \in X$. Then $p^{-1}(x), p^{-1}(y)$ are two finite sets of $\bar{X}$ with empty intersection. We write

$$p^{-1}(x) = \{\bar{x}_i\}_{i=1}^n, \quad p^{-1}(y) = \{\bar{y}_i\}_{i=1}^n$$

For each $i = 1, \dots, n$, let $U_i \ni \bar{x}_i, V_i \ni \bar{y}_i, U_i \cap V_i = \emptyset$. Then $p(\bigcup_i U_i) \cap p(\bigcup_i V_i) = \emptyset, x \in p(\bigcup_i U_i), y \in p(\bigcup_i V_i)$.

□

Basic Homology Algebra

4.1 Basic definition

Definition 4.1.1

Consider a direct sum

$$C_\bullet := C_* := \bigoplus_{n \in \mathbb{Z}} C_n$$

of R -modules^a. Often we call C_* a **graded-module** with its n^{th} **graded component** C_n . Members of C_n are also called **homogeneous elements** of C_* of degree n .

1. A **R -module homomorphism** $f : C_* \rightarrow C'_*$ is called a graded homomorphism if there exists d such that $f(C_r) \subseteq C'_{r+d}$ for all r . We then call d the degree of f . We shall denote $f|_{C_r}$ by f_r , and often we may simply write f itself for f_r provided that there is no confusion.
2. By a **chain complex** (C_*, ∂) of R -modules, we mean a graded R -module C_* , together with an endomorphism $\partial := \partial_* : C_* \rightarrow C_*$ of degree -1 with the property $\partial \circ \partial = 0$. The endomorphism ∂ is called the **differential** or the **boundry map** of the chain complex. Often we shall not mention the ∂ at all and merely say C_* is a chain complex.
3. If C and C_* are two chain complexes then by a **chain map** $f = f_* : C_* \rightarrow C'_*$ we mean a graded module homomorphism of degree 0 that commutes with the corresponding differentials.

^ais also a R -module

Remark.

1. The direct sum C_* is also an R -module.
2. Observe that ∂ consists of a sequence $\{\partial_n : C_n \rightarrow C_{n-1}\}$ of R -module homomorphisms such that $\partial_n \circ \partial_{n-1} = 0$ for all n .
3. f consists of a sequence $\{f_n : C_n \rightarrow C'_n\}$ of R -module homomorphisms such that $\partial'_n \circ f_n = f_{n-1} \circ \partial_n$ for all n . Expressed with an

diagram, that is

$$\begin{array}{ccc} C_n & \xrightarrow{\partial_n} & C_{n-1} \\ \downarrow f_n & & \downarrow f_{n-1} \\ C'_n & \xrightarrow{\partial'_n} & C'_{n-1} \end{array}$$

Proposition 4.1.2

There is a category of chain complexes of R -modules and chain maps. We shall denote this category by $\mathbf{Ch}_R$.

Proof. The objects are chain complexes. The morphisms are chain maps. For composition of morphisms, consider two chain maps $f : C_* \rightarrow C'_*$, $g : C'_* \rightarrow C''_*$. Only need to show that the following diagram commutes

$$\begin{array}{ccc} C_n & \xrightarrow{\partial_n} & C_{n-1} \\ \downarrow f_n & & \downarrow f_{n-1} \\ C'_n & \xrightarrow{\partial'_n} & C'_{n-1} \\ \downarrow f'_n & & \downarrow f'_{n-1} \\ C''_n & \xrightarrow{\partial''_n} & C''_{n-1} \end{array}$$

which is obvious. Finally, the existence of the identity map is also obvious. □

Definition 4.1.3 ▶ Direct Sum of Chain Complexes

Define the direct sum of a family of chain complexes $\{(C^\alpha, \partial^\alpha)\}_{\alpha \in \Lambda}$ as the chain complexes $(C, \partial) := (\oplus_\alpha C^\alpha, \oplus_\alpha \partial^\alpha)$, where the n^{th} graded component of C is

$$C_n = (C_n^\alpha)_{\alpha \in \Lambda},$$

∂ is defined as

$$\partial((c^\alpha)_{\alpha \in \Lambda}) := (\partial^\alpha(c^\alpha))_{\alpha \in \Lambda}$$

4.2 Exact Sequence

Definition 4.2.1 ► Exact Sequence

1. A sequence of R -modules

$$M' \xrightarrow{\alpha} M \xrightarrow{\beta} M''$$

is said to be **exact** at M if $\ker \beta = \operatorname{Im} \alpha$.

2. A sequence

$$\cdots \longrightarrow M_{n-1} \longrightarrow M_n \longrightarrow M_{n+1} \longrightarrow \cdots$$

is said to be exact if it is exact at each M_n .

3. By a short exact sequence we mean an exact sequence of the form

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

Definition 4.2.2 ► Exact Sequence of Chain Complexes

A sequence of chain complexes and chain maps

$$0 \longrightarrow C' \xrightarrow{f} C \xrightarrow{g} C'' \longrightarrow 0$$

is said to be exact if for each n the corresponding sequence of modules

$$0 \longrightarrow C'_n \xrightarrow{f_n} C_n \xrightarrow{g_n} C''_n \longrightarrow 0$$

is exact.

Lemma 4.2.3 ► Snake lemma

Given a commutative diagram of R -module homomorphisms: where the two horizontal sequences are exact, there exists a R -module homomorphism

$$\begin{array}{ccccccc} M' & \xrightarrow{\alpha} & M & \xrightarrow{\beta} & M'' & \longrightarrow & 0 \\ \downarrow f' & & \downarrow f & & \downarrow f'' & & \\ 0 & \longrightarrow & N' & \xrightarrow{\alpha'} & N & \xrightarrow{\beta'} & N'' \end{array}$$

$\delta : \operatorname{Ker} f'' \longrightarrow \operatorname{Coker} f'$, called the *connecting homomorphism* such that the sequence

$$\operatorname{Ker} f' \xrightarrow{\bar{\alpha}} \operatorname{Ker} f \xrightarrow{\bar{\beta}} \operatorname{Ker} f'' \xrightarrow{\delta} \operatorname{Coker} f' \xrightarrow{\bar{\alpha}'} \operatorname{Coker} f \xrightarrow{\bar{\beta}'} \operatorname{Coker} f''$$

is exact. Moreover, the **connecting homomorphism** δ has the naturality properties, so that the above assignment of a ‘snake’ to the corresponding ‘six-term’ exact sequence of modules defines a covariant functor.

Note.

最关键的信息是 **connecting homomorphism** δ . 构造的过程就是借助 β 的是 surjective, α' 是 injective, 来调整使得 β 和 α' 对应的箭头是可以反转的. 事实上在应用时, 这个 connecting homomorphism 的具体构造往往比它的存在性更重要.

Proof. to be finished We need to show

1. $\bar{\alpha}, \bar{\beta}$ are well defined, where $\bar{\alpha} := \alpha|_{\ker f'}, \bar{\beta} := \beta|_{\ker f}$
2. $\text{Im } \bar{\alpha} = \ker \bar{\beta}$.
3. $\bar{\alpha}', \bar{\beta}'$ are well defined, where $\bar{\alpha}'([x]) := [\alpha'(x)], \bar{\beta}'([y]) := [\beta'(y)]$
4. $\text{Im } \bar{\alpha}' = \ker \bar{\beta}'$
5. Construct a δ .
6. $\text{Im } \bar{\beta} = \ker \delta$
7. $\text{Im } \delta = \ker \bar{\alpha}'$

We prove the above points below.

1. By commutative, $f \circ \alpha(\ker f') = \alpha' \circ f'(\ker f') = 0$, which shows that $\alpha(\ker f') \subseteq \ker f$. Thus $\bar{\alpha}$ is well defined. $\bar{\beta}$'s is similar.
- 2.

$$\text{Im } \bar{\alpha} = \alpha(\ker f') \subseteq \text{Im } \alpha = \ker \beta \implies \beta(\text{Im } \bar{\alpha}) = 0 \implies \text{Im } \bar{\alpha} \subseteq \ker \bar{\beta}$$

The other side,

$$\ker \bar{\beta} = \ker \beta \cap \ker f = \text{Im } \alpha \cap \ker f$$

Take $y \in \ker \bar{\beta}$, there exists $x \in M'$ such that $y = \alpha(x)$ and $f(y) = f(\alpha(x)) = 0$. Since $f \circ \alpha = \alpha' \circ f'$, $\alpha' \circ f'(x) = 0$. Since α' is injective, $x \in \ker f'$. Thus $y \in \text{Im } \bar{\alpha}$.

3. For $\bar{\alpha}'$, we need to show that for $x_1, x_2 \in N'$ such that $x_1 - x_2 \in \text{Im } f'$, $\alpha'(x_1) - \alpha'(x_2) \in \text{Im } f$. It is true by commutative

$$\alpha'(x_1) - \alpha'(x_2) = \alpha'(x_1 - x_2) \in \text{Im } (\alpha' \circ f') = \text{Im } (f \circ \alpha) \subseteq \text{Im } f$$

$\bar{\beta}'$'s is similar.

4. Take $[y]_{\sim \text{Im } f} \in \text{Im } \bar{\alpha}'$, then there exists $[x]_{\sim \text{Im } f'} \in \text{Coker } f'$, such that $\alpha'([x]_{\sim \text{Im } f'}) = [y]_{\sim \text{Im } f}$. Then $\alpha'(x) - y \in \text{Im } f$, there exists $z \in M$, such that

$$\alpha'(x) - y = f(z)$$

By commutative and exactness: $\ker \beta' = \text{Im } \alpha'$,

$$\beta'(y) = \beta'(\alpha'(x) - f(z)) = 0 - \beta' \circ f(z) = -f'' \circ \beta(z) \in \text{Im } f''$$

, which shows that $\bar{\beta}'([y]_{\sim \text{Im } f}) = [0]_{\sim \text{Im } f''}$, $\text{Im } \bar{\alpha}' \subseteq \ker \bar{\beta}'$.

Then, take $[y]_{\sim \text{Im } f} \in \ker \bar{\beta}'$. Then $y \in \ker \beta' = \text{Im } \alpha'$. There exists $x \in N'$ such that $y = \alpha'(x)$. Then $[y]_{\sim \text{Im } f} = \alpha'([x]_{\sim \text{Im } f'})$, $[y]_{\sim \text{Im } f} \in \text{Im } \bar{\alpha}'$, $\ker \bar{\beta}' \subseteq \text{Im } \bar{\alpha}'$.

5. Take $z \in \ker f''$. Since β is surjective, there exists $y \in M$ such that $\beta(y) = z$. Once we show that $f(y) \in \text{Im } \alpha'$, we can define

$$\delta(z) = [(\alpha')^{-1}(f(y))]_{\sim \text{Im } f'}$$

provided that this definition is well defined. By commutative

$$\beta'(f(y)) = f''(\beta(y)) = f''(z) = 0.$$

Hence $f(y) \in \ker \beta' = \text{Im } \alpha'$. Finally, to show that it is well defined, we take $z_1, z_2 \in \ker f''$ such that $\beta(y_1) = \beta(y_2) = z$. We need to show that $(\alpha')^{-1}(f(y_1 - y_2)) \in \text{Im } f'$. Since $\beta(y_1) - \beta(y_2) = 0$, $\beta(y_1 - y_2) = 0$, $y_1 - y_2 \in \ker \beta = \text{Im } \alpha$. Suppose $\alpha(x) = y_1 - y_2$. Then

$$(\alpha')^{-1}(f(y_1 - y_2)) = (\alpha')^{-1}(f\alpha(x)) = (\alpha')^{-1}(\alpha' \circ f')(x) = f'(x) \in \text{Im } f'$$

, which completes the proof. □

Corollary 4.2.4

Consider the following commutative diagram of R -modules and R -linear maps in which the two rows are exact. If f_1 and f_3 are isomorphisms then so is f_2 .

$$\begin{array}{ccccccc}
0 & \longrightarrow & M_1 & \xrightarrow{\alpha_1} & M_2 & \xrightarrow{\alpha_2} & M_3 \longrightarrow 0 \\
& & \downarrow f_1 & & \downarrow f_2 & & \downarrow f_3 \\
0 & \longrightarrow & N_1 & \xrightarrow{\beta_1} & N_2 & \xrightarrow{\beta_2} & N_3 \longrightarrow 0
\end{array}$$

Proof. By snake lemma, we have

$$0 \xrightarrow{\alpha_1} \ker f_2 \xrightarrow{\alpha_2} 0 \xrightarrow{\delta} 0 \xrightarrow{\beta_1} \text{Coker } f_2 \xrightarrow{\beta_2} 0$$

is exact. Then we have

$$\begin{aligned}
0 &= \ker \alpha_2 = \text{Im } \alpha_1 = 0 \\
&= \ker f_2
\end{aligned}$$

which shows that $\ker f_2 = 0$. Similarly, $\text{Coker } f_2 = 0$. The above shows that f_2 is an isomorphism. $\square$

Corollary 4.2.5 ▶ Four lemma

Consider the following commutative diagram of R modules and R -linear maps in which the two rows are exact. Suppose that f_1 is surjective and f_4 is injective. Then

- (i) f_2 is injective $\implies f_3$ is injective.
- (ii) f_3 is surjective $\implies f_2$ is surjective.

$$\begin{array}{ccccccc}
M_1 & \xrightarrow{\alpha_1} & M_2 & \xrightarrow{\alpha_2} & M_3 & \xrightarrow{\alpha_3} & M_4 \\
\downarrow f_1 & & \downarrow f_2 & & \downarrow f_3 & & \downarrow f_4 \\
N_1 & \xrightarrow{\beta_1} & N_2 & \xrightarrow{\beta_2} & N_3 & \xrightarrow{\beta_3} & N_4
\end{array}$$

Proof sketch.

如果 f_3 被两个单射 f_2, f_4 夹在中间, 得益于 f_1 是满射, 可以将右三列保持单射性地调整为一条 snake, 从而导出 f_3 是单射.

类似地, 如果 f_2 被两个满射 f_1, f_3 夹在中间, 得益于 f_4 是单射, 左三列可以保持满射性地调整为一条 snake, 导出 f_2 是满射.

Proof. 1. We can construct a snake

$$\begin{array}{ccccccc}
 M_2/\ker \alpha_2 & \xrightarrow{\alpha_2} & M_3 & \xrightarrow{\alpha_3} & \operatorname{Im} \alpha_3 & \longrightarrow & 0 \\
 \downarrow \bar{f}_2 & & \downarrow f_3 & & \downarrow f_4|_{\operatorname{Im} \alpha_3} & & \\
 0 \longrightarrow & N_2/\ker \beta_2 & \xrightarrow{\beta_2} & N_3 & \xrightarrow{\beta_3} & N_4 &
 \end{array}$$

By snake lemma, the following sequence is exact

$$\ker \bar{f}_2 \rightarrow \ker f_3 \rightarrow \ker f_4|_{\operatorname{Im} \alpha_3}$$

Since f_4 is injective, $f_4|_{\operatorname{Im} \alpha_3}$ is as well. Furthermore, by applying the snake lemma to the following diagram

$$\begin{array}{ccccccc}
 M_1 & \xrightarrow{\alpha_1} & M_2 & \xrightarrow{p} & M_2/\ker \alpha_2 & \longrightarrow & 0 \\
 \bar{f}_1 \downarrow & & \downarrow f_2 & & \downarrow \bar{f}_2 & & \\
 0 \longrightarrow & N_1/\ker \beta_1 & \xrightarrow{\bar{\beta}_1} & N_2 & \xrightarrow{p} & N_2/\ker \beta_2 &
 \end{array}$$

We get

$$0 = \ker f_2 \rightarrow \ker \bar{f}_2 \xrightarrow{\delta} \operatorname{Coker} \bar{f}_1$$

is a exact suquence,where $\operatorname{Coker} \bar{f}_1 = 0$ since f_1 is surjective and $\bar{f}_1$ is as well. It shows that $\ker \bar{f}_2 = 0$. Finally, we have the following sequence is exact

$$0 = \ker \bar{f}_2 \rightarrow \ker f_3 \rightarrow \ker f_4|_{\operatorname{Im} \alpha_3} = 0$$

It follows that $\ker f_3 = 0$, f_3 is injective.

2. Another snake we can construct is

$$\begin{array}{ccccccc}
 M_1 & \xrightarrow{\alpha_1} & M_2 & \xrightarrow{\alpha_2} & \operatorname{Im} \alpha_2 = \ker \alpha_3 & \longrightarrow & 0 \\
 \downarrow \bar{f}_1 & & \downarrow f_2 & & \downarrow f_3|_{\operatorname{Im} \alpha_2} & & \\
 0 \longrightarrow & N_1/\ker \beta_1 & \xrightarrow{\bar{\beta}_1} & N_2 & \xrightarrow{\beta_2} & \operatorname{Im} \beta_2 = \ker \beta_3 &
 \end{array}$$

Then by snake lemma ,the following sequence is exact

$$\ker \bar{f}_1 \rightarrow \ker f_2 \rightarrow \ker f_3|_{\operatorname{Im} \alpha_2} \rightarrow \operatorname{Coker} \bar{f}_1 \rightarrow \operatorname{Coker} f_2 \rightarrow \operatorname{Coker} f_3|_{\operatorname{Im} \alpha_2}$$

Furthermore, by applying the snake lemma to the following diagram

$$\begin{array}{ccccccccc}
 0 & \longrightarrow & \text{Ker } \alpha_3 & \xrightarrow{\iota_M} & M_3 & \xrightarrow{\alpha_3} & \text{Im } \alpha_3 & \longrightarrow & 0 \\
 & & \downarrow f_3|_{\text{Im } \alpha_2} & & \downarrow f_3 & & \downarrow f_4|_{\text{Im } \alpha_3} & & \\
 0 & \longrightarrow & \text{Ker } \beta_3 & \xrightarrow{\iota_N} & N_3 & \xrightarrow{\beta_3} & \text{Im } \beta_3 & \longrightarrow & 0
 \end{array}$$

We get

$$0 = \ker f_4|_{\text{Im } \alpha_3} \xrightarrow{\delta} \text{Coker}(f_3|_{\text{Im } \alpha_2}) \rightarrow \text{Coker } f_3 = 0$$

which shows that $\text{Coker}(f_3|_{\text{Im } \alpha_2}) = 0$. Finally, we have the following sequence exact

$$0 = \text{Coker } \bar{f}_1 \rightarrow \text{Coker } f_2 \rightarrow \text{Coker } f_3|_{\text{Im } \alpha_2} = 0$$

Thus $\text{Coker } f_2 = 0$, f_2 is surjective.

□

Corollary 4.2.6 ▶ Five lemma

In the following diagram of R -modules, the two rows are given to be exact. If f_1, f_2, f_4 and f_5 are isomorphisms then f_3 is also an isomorphism.

$$\begin{array}{ccccccccc}
 M_1 & \longrightarrow & M_2 & \longrightarrow & M_3 & \longrightarrow & M_4 & \longrightarrow & M_5 \\
 \downarrow f_1 & & \downarrow f_2 & & \downarrow f_3 & & \downarrow f_4 & & \downarrow f_5 \\
 M'_1 & \longrightarrow & M'_2 & \longrightarrow & M'_3 & \longrightarrow & M'_4 & \longrightarrow & M'_5
 \end{array}$$

Proof. By applying the Four lemma to the left four columns, we get f_3 is injective. And by applying the Four lemma to the right four columns, we get f_3 is surjective.

□

4.3 Homology

Definition 4.3.1

Given a chain complex C_* , define the **homology group** of C_* to be the graded R -module

$$H_*(C_*) := \bigoplus_{n \in \mathbb{Z}} H_n(C_*)$$

by taking

$$H_n(C_*) := \text{Ker } \partial_n / \text{Im } \partial_{n+1}, \quad \forall n \in \mathbb{Z}.$$

Proposition 4.3.2

If $f : C_* \rightarrow C'_*$ is a chain map then f induces a graded homomorphism

$$H_*(f) : H_*(C_*) \rightarrow H_*(C'_*)$$

In addition, this has the naturality property, viz.,

1. $H_*(\text{Id}) = \text{Id}$
2. If g is another chain map such that $f \circ g$ is defined, then

$$H_*(f \circ g) = H_*(f) \circ H_*(g)$$

Thus, H_* is a **covariant functor** from the category of chain complexes to the category of graded modules.

Theorem 4.3.3

The homology of a direct sum of chain complexes is isomorphic to the direct sum of the homology of chain complexes.

Proof.

$$H_n(\bigoplus_{\alpha} C_*^{\alpha}) = \text{ker}(\bigoplus_{\alpha} \partial_n^{\alpha}) / \text{Im}(\bigoplus_{\alpha} \partial_{n+1}^{\alpha})$$

$$\bigoplus_{\alpha} H_n(C_*^{\alpha}) = \bigoplus_{\alpha} (\text{ker } \partial_n^{\alpha} / \text{Im } \partial_{n+1}^{\alpha})$$

There are obvious isomorphisms :

$$\text{ker}(\bigoplus_{\alpha} \partial_n^{\alpha}) \simeq \bigoplus_{\alpha} \text{ker } \partial_n^{\alpha}, \quad \text{Im}(\bigoplus_{\alpha} \partial_{n+1}^{\alpha}) \simeq \bigoplus_{\alpha} \text{Im } \partial_{n+1}^{\alpha}$$

A standard result in algebra is that if K_α is a submodule of M_α , then

$$\frac{\bigoplus_\alpha M_\alpha}{\bigoplus_\alpha K_\alpha} \cong \bigoplus_\alpha \frac{M_\alpha}{K_\alpha}$$

. Hence , we have

$$\bigoplus_\alpha \ker \partial_n^\alpha / \bigoplus_\alpha \operatorname{Im} \partial_{n+1}^\alpha \simeq \bigoplus_\alpha (\ker \partial_n^\alpha / \operatorname{Im} \partial_{n+1}^\alpha)$$

□

Theorem 4.3.4

Given a short exact sequence of chain complexes

$$0 \rightarrow C'_* \xrightarrow{\alpha} C_* \xrightarrow{\beta} C''_* \rightarrow 0$$

there is a functorial long exact sequence of homology groups

$$\rightarrow H_n(C'_*) \xrightarrow{H_n(\alpha)} H_n(C_*) \xrightarrow{H_n(\beta)} H_n(C''_*) \xrightarrow{\delta_n} H_{n-1}(C'_*) \xrightarrow{H_{n-1}(\alpha)} H_{n-1}(C_*) \rightarrow$$

Proof sketch.

Consider the diagram

$$\begin{array}{ccccccc} C'_n / \operatorname{Im} \partial'_{n+1} & \xrightarrow{\bar{\alpha}_n} & C_n / \operatorname{Im} \partial_{n+1} & \xrightarrow{\bar{\beta}_n} & C''_n / \operatorname{Im} \partial''_{n+1} & \longrightarrow & 0 \\ \downarrow \partial'_n & & \downarrow \partial_n & & \downarrow \partial''_{n+1} & & \\ 0 \longrightarrow & \operatorname{Ker} \partial'_{n-1} & \xrightarrow{\alpha'_{n-1}} & \operatorname{Ker} \partial_{n-1} & \xrightarrow{\beta'_{n-1}} & \operatorname{Ker} \partial''_{n-1} & \end{array}$$

Singular Homology

5.1 The Construction

Definition 5.1.1

Let X be any topological space and $n \geq 0$ be an integer. By a **singular n -simplex** in X , we mean a continuous map $\sigma : |\Delta_n| \rightarrow X$. By a **singular n -chain** in X , we mean a finite sum $\sum_i n_i \sigma_i$, where $n_i \in \mathbb{Z}$ and σ_i are singular n -simplices in X . We can add any two chains in an obvious way to obtain another chain by simply following the rule

$$n\sigma + m\sigma = (n + m)\sigma \quad (4.5)$$

Then the collection of all singular n -chains in X becomes a free abelian group, with the empty sum playing the role of the zero-element. The collection of all singular n -simplices is then a basis for this group. We denote this group by $S_n(X)$.

Proposition 5.1.2 ► Functoriality

If $f : X \rightarrow Y$ is a map of topological spaces, then $\sigma \mapsto f \circ \sigma$ extends to define a graded group homomorphism $f_* : S_*(X) \rightarrow S_*(Y)$. Since $(g \circ f) \circ \sigma = g \circ (f \circ \sigma)$, it follows that the association $X \mapsto S_*(X)$ defines a functor.

Definition 5.1.3 ► Relative Chain Complex

If A is a subspace of X and σ is a singular simplex in A then it can be considered as a singular simplex in X via the inclusion map $A \hookrightarrow X$. In this way, $S_n(A)$ becomes a subgroup of $S_n(X)$. We denote the quotient group $S_n(X)/S_n(A)$ by $S_n(X, A)$.

Remark.

1. $S_n(X, A)$ is a free abelian group with the collection of all singular n -simplexes σ in X whose image is not contained in A as a basis.

2. If $A = \emptyset$, then clearly $S_n(X, A) = S_n(X)$.

Definition 5.1.4

For each integer $r \geq 0$, the **face map** or the **face operator** $F^r : \mathbb{R}^\infty \rightarrow \mathbb{R}^\infty$ is defined by

$$F^r(\mathbf{e}_s) = \begin{cases} \mathbf{e}_s, & \text{if } s < r, \\ \mathbf{e}_{s+1}, & \text{if } s \geq r \end{cases}$$

and extended linearly.

Remark.

1. Note that for each $n \geq r$, F^r carries Δ_{n-1} onto the $(n-1)$ -face of Δ_n opposite to the vertex $\mathbf{e}_r$, in an **order preserving** manner.
2. We shall denote each F^r restricted to any $\Delta_n, n \geq r$ also by F^r , the exact meaning being understood from the context.

Lemma 5.1.5

$$F^r \circ F^{s-1} = F^s \circ F^r \text{ if } r < s. \text{ }^a$$

^a这里修正了 shastri 书中的一个笔误, 他可能写错了 $s-1$ 的位置

Proof. Since $r < s, r \leq s-1$, we fully expand the LHS by definition

$$\begin{aligned} F^r \circ F^{s-1}(\mathbf{e}_k) &= \begin{cases} F^r(\mathbf{e}_k), & k < s-1 \\ F^r(\mathbf{e}_{k+1}), & k \geq s-1 \end{cases} \\ &= \begin{cases} \mathbf{e}_k, & k < s-1, k < r \\ \mathbf{e}_{k+1}, & r \leq k < s-1 \\ \mathbf{e}_{k+1}, & s-1 \leq k < r-1 \text{ (empty set)} \\ \mathbf{e}_{k+2}, & k \geq s-1, k \geq r-1 \end{cases} \\ &= \begin{cases} \mathbf{e}_k, & k < r \\ \mathbf{e}_{k+1}, & r \leq k < s-1 \\ \mathbf{e}_{k+2}, & k \geq s-1 \end{cases} \end{aligned}$$

Similarly,

$$\begin{aligned}
 F^s \circ F^r (\mathbf{e}_k) &= \begin{cases} F^s (\mathbf{e}_k), & k < r \\ F^s (\mathbf{e}_{k+1}), & k \geq r \end{cases} \\
 &= \begin{cases} \mathbf{e}_k, & k < r, k \leq s \\ \mathbf{e}_{k+1}, & k < r, k \geq s \text{ (empty set)} \\ \mathbf{e}_{k+1}, & k \geq r, k < s-1 \\ \mathbf{e}_{k+2}, & k \geq r, k \geq s-1 \end{cases} \\
 &= \begin{cases} \mathbf{e}_k, & k < r \\ \mathbf{e}_{k+1}, & r \leq k < s-1 \\ \mathbf{e}_{k+2}, & k \geq s-1 \end{cases}
 \end{aligned}$$

□

Definition 5.1.6

For $n \geq 1$ and for any singular n -simplex σ in X , we define

$$\partial_n(\sigma) = \sum_{r=0}^n (-1)^r \sigma \circ F^r$$

and extend it linearly to obtain homomorphisms $\partial_n : S_n(X, A) \rightarrow S_{n-1}(X, A)$. We define $S_n(X, A) = (0)$ for $n \leq -1$ and $\partial_n = 0$ for $n \geq 0$.

Note.

σ is a map from Δ_n to X . $\sigma \circ F^r$ is the restriction of σ on its face opposite to $\mathbf{e}_r$. Then $\partial_n(\sigma)$ is something like a closed $(n-1)$ -surface consists of σ 's faces.

Proposition 5.1.7 ► Singular Chain Complex

$(S(X, A), \partial) = \{S_n(X, A), \partial_n\}$ is a chain complex and the association $(X, A) \rightsquigarrow (S(X, A), \partial)$ is a functor.

Proof. We shall first show that $\partial^2(\sigma) = 0$ for any singular n -simplex, $n \geq 2$:

$$\begin{aligned}
 \partial^2(\sigma) &= \partial \left(\sum_{s=0}^n (-1)^s \sigma \circ F^s \right) \\
 &= \sum_{s=0}^n (-1)^s \partial(\sigma \circ F^s) \\
 &= \sum_{s=0}^n \sum_{r=0}^{n-1} (-1)^s (-1)^r \sigma \circ F^s \circ F^r \\
 &= \sum_{0 \leq r < s \leq n} (-1)^{s+r} \sigma \circ F^r \circ F^{s-1} + \sum_{n-1 \geq r \geq s \geq 0} (-1)^{s+r} \sigma \circ F^s \circ F^r \\
 &= \sum_{0 \leq r < s \leq n} (-1)^{s+r} \sigma \circ F^r \circ F^{s-1} + \sum_{n-1 \geq s \geq r \geq 0} (-1)^{s+r} \sigma \circ F^r \circ F^s \\
 &= \sum_{0 \leq r < s \leq n} (-1)^{s+r} \sigma \circ F^r \circ F^{s-1} - \sum_{n \geq s > r \geq 0} (-1)^{s+r} \sigma \circ F^r \circ F^{s-1} \\
 &= 0
 \end{aligned}$$

Given a map $f : (X, A) \rightarrow (Y, B)$. Since $(f \circ \sigma) \circ F^r = f \circ (\sigma \circ F^r)$, we define $f_*(\sigma) := f \circ \sigma$ extended by linearity. Then $\partial(f_*(\sigma)) = f_*(\partial(\sigma))$ by the definition of ∂ and f_* 's linearity. The rest of the details are omitted. $\square$

Definition 5.1.8

The **relative singular homology groups** of a topological pair (X, A) are the homology groups $H_*(S_*(X, A))$ and are denoted by $H_*(X, A)$. Taking $A = \emptyset$, we get the homology groups $H_*(X)$.

Remark.

Given any commutative ring R

1. We could have formed the **free module** $S_n(X; R)$ of all the singular n -chains in X .
2. Exactly as before, we would have got a chain complex $S_*(X; R)$, sub-chain complex $S_*(A; R)$ and the quotient complex $S_*(X, A; R)$, etc.
3. The corresponding homology groups $H_*(X; R)$, etc., are then called the homology groups X with coefficients in R . Clearly they form a **graded R -module** by themselves. It is customary to drop the notation R when $R = \mathbb{Z}$.

5.2 Basic Properties

Remark.

- It is easily seen that $\{S_*(X, A), \partial\}$ is a functor from the category of topological pairs to the category of chain complexes; thus $H_*(X, A)$ and $H_*(X)$ are also functors.
- Given $f : (X, A) \rightarrow (Y, B)$, we shall denote by f_* , the map induced by f on $S_*(X, A)$ and by f_* the map induced by f on $H_*(X, A)$.

Lemma 5.2.1

There exist functorial chain homotopies

$$h : S_q(X, A) \rightarrow S_{q+1}((X, A) \times \mathbb{I}), q \geq 0,$$

,called **prism operator**, between the two inclusion induced maps $\eta_0, \eta_1 : S_*(X, A) \rightarrow S_*((X, A) \times \mathbb{I})$, where for any $t \in \mathbb{I}$, η_t is the inclusion map

$$\eta_t(x) = (x, t), \quad t \in \mathbb{I}$$

Theorem 5.2.2 ▶ Homotopy invariance

Let $f, g : (X, A) \rightarrow (Y, B)$ be **homotopic** to **each other**. Then $f_* = g_*$ on $H_*(X, A)$.

Proof. Let $G : (X, A) \times \mathbb{I} \rightarrow (Y, B)$ be a homotopy between f and g . For any $t \in \mathbb{I}$, let $\eta_t : (X, A) \rightarrow (X, A) \times \mathbb{I}$ be inclusion map $\eta_t(x) = (x, t)$. Then $G \circ \eta_0 = f$, $G \circ \eta_1 = g$. Only need to show that chain maps $(\eta_0)_*$ and $(\eta_1)_*$ are chain homotopic, thus $(\eta_0)_* = (\eta_1)_*$, $f_* = g_*$, which is implied by the lemma above. $\square$

Corollary 5.2.3

Homotopy-equivalent topological spaces have isomorphic homology groups; in particular, a contractible space has the homology groups of a point-space.

Example 5.2.4

Let $\star$ be a point space. We have

$$H_k(\star) = \begin{cases} \mathbb{Z}, & k = 0 \\ 0, & k \geq 1 \end{cases}$$

Proof. Let n be any positive integer. There is a unique singular n -simplex $\sigma : |\Delta_n| \rightarrow \{\star\}$ and hence $S_n(\star) \cong \mathbb{Z}$ for each $n \geq 0$. Observe that $\partial : S_{2n+1}(\star) \rightarrow S_{2n}(\star)$ is the zero-map because Δ_{2n+1} has an even number of $(2n)$ -faces and the terms in $\partial(\sigma)$ cancel in pairs. However, similar reasoning tells us that $\partial : S_{2n}(\star) \rightarrow S_{2n-1}(\star)$ is an isomorphism. Thus we have a chain complex

$$\dots \xrightarrow{0} \mathbb{Z} \xrightarrow{\cong} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{\cong} \mathbb{Z} \longrightarrow 0$$

Thus it follows that $H_0(\star) \cong \mathbb{Z}$ and $H_i(\star) = (0)$ for all $i > 0$. $\square$

Example 5.2.5

Let $\{X_j : j \in J\}$ be the set of path connected components of X so that $X = \bigsqcup_{j \in J} X_j$. Then

$$H_*(X) = \bigoplus_{j \in J} H_*(X_j)$$

Remark.

Thus in practice, we can often assume that the space itself is path connected, while dealing with its homology groups.

Proof. Since $|\Delta_n|$ is a path connected space, it follows that any singular n -simplex in X is contained completely in one of the path components X_j . Therefore, it follows that the entire chain complex $S_*(X)$ is the direct sum of the subchain complexes $S_*(X_j)$. Hence the homology is also a direct sum of $H_*(X_j)$, $j \in J$ by Theorem 4.3.3. $\square$

Theorem 5.2.6 ▶ Homology Long Exact Sequence of the pair (X, A)

By definition of $S_*(X, A)$, we have an exact sequence of chain complexes

$$0 \rightarrow S_*(A) \rightarrow S_*(X) \rightarrow S_*(X, A) \rightarrow 0.$$

By Theorem 4.3.4, this yields the long exact sequence of homology groups

$$\begin{cases} \dots \rightarrow H_i(A) \rightarrow H_i(X) \rightarrow H_i(X, A) \xrightarrow{\delta} H_{i-1}(A) \rightarrow \dots \\ \dots \rightarrow H_1(X, A) \xrightarrow{\delta} H_0(A) \rightarrow H_0(X) \rightarrow H_0(X, A), \end{cases} \quad (4.6)$$

which is functorial.

Remark.

This sequence comes handy in many computations as we shall see soon.

5.3 Homology Excision and Mayer-Vietoris

Lemma 5.3.1

Let $X = A \cup B$. Then the following statements are equivalent.

1. $S_*(A) + S_*(B) \rightarrow S_*(X)$ induces isomorphisms in homology.
2. $[S_*(A) + S_*(B)]/S_*(B) \rightarrow S_*(X)/S_*(B)$ induces isomorphisms in homology.
3. $[S_*(A) + S_*(B)]/S_*(A) \rightarrow S_*(X)/S_*(A)$ induces isomorphisms in homology.
4. $S_*(A)/S_*(A \cap B) \rightarrow S_*(X)/S_*(B)$ induces isomorphisms in homology.
5. $S_*(B)/S_*(A \cap B) \rightarrow S_*(X)/S_*(A)$ induces isomorphisms in homology.

Note.

1,2 等价的好处就在于, 它告诉我们, 我们所需要的同调群的同构对于一部分的信息是不敏感的。

Proof sketch.

对于 1,2 的等价性, 涉及四种同调群, 我们借助 B 的同调将它们相互联系. 用短正合列到长正合列的诱导行为将四种同调群两两联系, 并以 B 的同调为桥梁, 利用诱导行为的函子性将两组同调群联系起来. B 到自身的恒等给出一组同构, 配合

已知的一组同构可以利用 Five lemma 夹出另一组同构.
2,4 的等价性直接就是第二同构定理的结果, 本质上与同调关系不大.

Proof. The equivalence of (1) and (2) is a consequence of the Five lemma applied to the ladder of homology long exact sequences given by the ladder of short exact sequences

$$\begin{array}{ccccccc}
 0 & \longrightarrow & S.(B) & \longrightarrow & S.(A) + S.(B) & \longrightarrow & [S.(A) + S.(B)]/S.(B) \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & S.(B) & \longrightarrow & S.(X) & \longrightarrow & S.(X)/S.(B) \longrightarrow 0.
 \end{array}$$

The equivalence of (2) and (4) follows from the commutative diagram below in which the horizontal arrow is the isomorphism given by the Noether's isomorphism theorem.

$$\begin{array}{ccc}
 S.(A)/S.(A \cap B) & \longrightarrow & [S.(A) + S.(B)]/S.(B) \\
 & \searrow & \downarrow \\
 & & S.(X)/S.(B)
 \end{array}$$

□

Definition 5.3.2

Let X be a topological space, A, B be two subspaces such that $X = A \cup B$. We then say that the inclusion map $(A, A \cap B) \hookrightarrow (X, B)$ an excision map. A pair $\{A, B\}$ of subspaces of a space X is called an **excisive couple** for singular homology, if the inclusion map

$$S.(A) + S.(B) \hookrightarrow S.(A \cup B)$$

induces isomorphisms in homology or if any of the other four equivalent conditions in the above lemma is satisfied.

Note.

excisive couple 是一对能代表 (X, A) 和 (X, B) 的同调性质的子集 (A, B) , 是 X 的一个良好分解. 它使得全空间 X 在切除 $X \setminus A$ 或 $X \setminus B$ 的切除行为上, 在同调的意义下是无损的.

Theorem 5.3.3 ▶ Homology excision

The inclusion map $(A, A \cap B) \hookrightarrow (A \cup B, B)$ of an excisive couple $\{A, B\}$ induces an isomorphism of homology groups $H_*(A, A \cap B) \rightarrow H_*(A \cup B, B)$.

Theorem 5.3.4

If $X = X_1 \cup X_2 = \text{int}_X(X_1) \cup \text{int}_X(X_2)$ then, $\{X_1, X_2\}$ is an excisive couple for the singular homology.

Note.

也就是说, 当分解的一个组件的边界能被另一个组件的内部“吃掉”时, 这组分解就是切除对.

Example 5.3.5

Here are a few examples of excisive couples which occur in practice.

1. $\{\mathbb{D}^n, \mathbb{R}^n \setminus \{0\}\}$ is an excisive couple and hence the inclusion map $(\mathbb{D}^n, \mathbb{D}^n \setminus \{0\}) \hookrightarrow (\mathbb{R}^n, \mathbb{R}^n \setminus \{0\})$ will induce isomorphisms in all singular homology groups.
2. Let N denote the north pole on $\mathbb{S}^n$ and $\bar{U}$ denote the closed upper hemisphere. Then $\{\mathbb{S}^n \setminus \{N\}, \bar{U}\}$ is an excisive couple, which gives isomorphisms of $H_*(\bar{U}, \bar{U} \setminus \{N\}) \rightarrow H_*(\mathbb{S}^n, \mathbb{S}^n \setminus \{N\})$.

Theorem 5.3.6

Given subsets X_1 and X_2 of a topological space X , let us denote the inclusion maps of X_i in to X by $\eta_i, i = 1, 2$. They induce inclusion maps of the chain complexes which we shall denote again by $\eta_i : S.(X_i) \rightarrow S.(X), i = 1, 2$. Now consider the chain map

$$(\eta_1, -\eta_2) : S.(X_1) \oplus S.(X_2) \rightarrow S.(X).$$

The submodule generated by $S.(X_1)$ and $S.(X_2)$ in $S.(X)$ is denoted by $S.(X_1) + S.(X_2)$. Then we have a short exact sequence of chain complexes

$$0 \rightarrow S.(X_1 \cap X_2) \rightarrow S.(X_1) \oplus S.(X_2) \rightarrow S.(X_1) + S.(X_2) \rightarrow 0$$

If $\{X_1, X_2\}$ is an excisive couple, then we can obtain the **Mayer-Vietoris**

sequence:

$$\left. \begin{aligned} \cdots \rightarrow H_{i+1}(X_1 \cup X_2) \xrightarrow{\partial} H_i(X_1 \cap X_2) \xrightarrow{i_*} H_i(X_1) \oplus H_i(X_2) \xrightarrow{j_*} H_i(X_1 \cup X_2) \rightarrow \cdots \\ \cdots \rightarrow H_1(X_1 \cup X_2) \xrightarrow{\partial} H_0(X_1 \cap X_2) \rightarrow H_0(X_1) \oplus H_0(X_2) \rightarrow H_0(X_1 \cup X_2) \end{aligned} \right\}$$

with

$$i_*(z) = (i_{1*}z, -i_{2*}z); \quad j_*(z_1, z_2) = j_{1*}z_1 + j_{2*}z_2$$

where,

$$\begin{aligned} i_1 : X_1 \cap X_2 \hookrightarrow X_1, \quad i_2 : X_1 \cap X_2 \hookrightarrow X_2 \\ j_1 : X_1 \hookrightarrow X_1 \cup X_2, \quad j_2 : X_2 \hookrightarrow X_1 \cup X_2 \end{aligned}$$

are inclusion maps.

Remark.

The Mayer-Vietoris sequence is one of the major ready-to-use-tools in computing homology.

Proof. The kernel of $(\eta_1, -\eta_2)$ is $S.(X_1) \cap S.(X_2) = S.(X_1 \cap X_2)$. The image is clearly $S.(X_1) + S.(X_2)$. Thus the short sequence is exact. Then we obtain the long exact sequence

$$\begin{aligned} \cdots \rightarrow H_{i+1}(S.(X_1) + S.(X_2)) \xrightarrow{\partial} H_i(X_1 \cap X_2) \xrightarrow{i_*} H_i(X_1) \oplus H_i(X_2) \xrightarrow{j_*} H_i(S.(X_1) + S.(X_2)) \\ \cdots \rightarrow H_1(S.(X_1) + S.(X_2)) \xrightarrow{\partial} H_0(X_1 \cap X_2) \rightarrow H_0(X_1) \oplus H_0(X_2) \rightarrow H_0(S.(X_1) + S.(X_2)) \end{aligned}$$

By the definition of excisive couple, we can replace $H_i(S.(X_1) + S.(X_2))$ by $H_i(X_1 \cup X_2)$ to obtain Mayer-Vietoris. $\square$

5.4 Problems

Problem 5.4.1

Compute the homology of $\mathbb{S}^p \times \mathbb{S}^q$ using Mayer-Vietoris.

Proof. We assume that $p, q \geq 1$. Let U^+, U^- be $\mathbb{S}^q$ without the north pole and south pole respectively. Then $(A, B) := (\mathbb{S}^p \times U^+, \mathbb{S}^p \times U^-)$ is a excisive

couple of $\mathbb{S}^p \times \mathbb{S}^q$. Since $U^+ \cap U^-$ can deformation retract to the equator of $\mathbb{S}^q$, which is $\mathbb{S}^{q-1}$, then $A \cap B$ can deformation retract to $\mathbb{S}^p \times \mathbb{S}^{q-1}$.

$$\cdots \xrightarrow{i'_*} H_{i+1}(\mathbb{S}^p) \oplus H_{i+1}(\mathbb{S}^q) \xrightarrow{j_*} H_{i+1}(\mathbb{S}^p \times \mathbb{S}^q) \xrightarrow{\partial} H_i(\mathbb{S}^p \times \mathbb{S}^{q-1}) \xrightarrow{i_*} H_i(\mathbb{S}^p) \oplus H_i(\mathbb{S}^q) -$$

The following short exact sequence separates

$$0 \rightarrow \ker \partial \xrightarrow{\iota} H_{i+1}(\mathbb{S}^p \times \mathbb{S}^q) \xrightarrow{\partial} \text{Im } \partial \rightarrow 0$$

which is equivalent to

$$0 \rightarrow \text{Im } j_* \simeq \text{Coker } j_* \xrightarrow{\iota} H_{i+1}(\mathbb{S}^p \times \mathbb{S}^q) \xrightarrow{\partial} \ker i_* \rightarrow 0$$

where $\text{Im } j_* \simeq H_{i+1}(\mathbb{S}^p) \oplus H_{i+1}(\mathbb{S}^q) / \text{Im } i'_* = \text{Coker } i'_*$. Thus we have

$$H_{i+1}(\mathbb{S}^p \times \mathbb{S}^q) \simeq \ker i_* \oplus \text{Coker } i'_*$$

$$\text{Im } i_* \simeq H_i$$

1. When $p = q$

(a)

□

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